Journal of Transport & Health 46 (2026) 102210

Contents lists available at ScienceDirect

Journal of Transport & Health

journal homepage: www.elsevier.com/locate/jth

Analyzing women's perceived safety in train commuting using the ABC model

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ARTICLE INFO

Keywords:
Public transport
Train systems
Women commuters
Commuting
Perceived safety
ABC model
Gender studies

ABSTRACT

Introduction: The right to safety is a basic need of people, and one of the biggest concerns on safety revolves around public transportation, particularly among women. This study examines factors influencing the perceived safety of women commuters in Metro Manila's train system.
Conceptual framework: Using Wirga and Debarnardi's ABC Model, the study identifies media exposure, familial upbringing, witnessing negative experiences, and experiencing negative experiences as predictors of perceived safety.
Method: A total of 200 women commuters who have used any train system in Metro Manila, Philippines at least once in the past year participated in an online survey that measured the factors in the conceptual model.
Results: Regression analysis revealed that Media Exposure, Familial Upbringing, and Experiencing Negative Experiences significantly predict Perceived Safety, while Witnessing Negative Experiences did not.
Implications: These findings highlight the critical role of Media Exposure, Familial Upbringing, and Experiencing Negative Experiences in shaping women's safety perceptions while commuting. Addressing these vulnerabilities can enhance the safety and inclusivity of public transportation for female commuters.

1. Introduction

The right to safety—defined as the "highest attainable standard of protection against natural and man-made hazards" (Twigg, 2003)—is one of the most fundamental rights of a person. As per Maslow's Hierarchy of Needs, it is a basic need that comes second only to physiological needs (Maslow, 1943). There is no denying that safety is crucial to the improvement of society, and that all individuals must feel safe in a space, or community. Thus, concerns on safety must be properly evaluated and studied thoroughly as communities work towards a safer environment. Women's safety, in particular, must be given special consideration as their safety in public spaces has historically been a concern (Valentine, 1989; Pain, 1997).

One of the biggest concerns on safety revolves around perceived safety in public transportation, particularly of those coming from

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https://doi.org/10.1016/j.jth.2025.102210
Received 21 July 2025; Received in revised form 14 October 2025; Accepted 5 November 2025
Available online 10 November 2025
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the female demographic. Fear of crime continues to be a widespread problem, and it tends to be more prominent amongst women (Day et al., 2003; as cited in Yavuz and Welch, 2010). In particular, train transit is often perceived as the most fear-inducing public activity (Walsh, 1999; Lynch and Atkins, 1988; as cited in Yavuz and Welch, 2010). The most common safety and security concerns of women in railway transit include instances of sexual harassment, theft, fraud, and aggression (Coppola and Silvestri, 2020).

Numerous studies on the perceived safety of females in public transportation have been conducted across the world. A study conducted in Chicago, USA by Nilay Yavuz and Eric Welch (2010) showed that compared to men, "women on average report lower levels of perceived train safety, a higher number of negative experiences and lower satisfaction with the presence of police and video cameras." The study of Bachok et al. (2014) in Jakarta focused on women-only coaches (WOC) in train systems, showing the lack of police enforcement, insufficient capacity to accommodate increasing demand, and the poor visual quality of the signs indicating and distinguishing WOC from the other rail-cars.

While numerous studies on the subject matter are present internationally, limited studies on female commuter safety have focused on the Philippines. This emphasizes a gap in the knowledge—there is limited information that specifically outlines factors that influence perceived safety among women in such contexts. Studying factors that influence perceived safety among women commuters is crucial, for it will help various stakeholders create bespoke safety measures attuned to the specific needs of women in such contexts. As such, this study intends to understand and determine the factors that influence the perceived safety of female commuters—defined to also include those who identify as a woman—who ride the train system in Metro Manila, Philippines.

Wirga and DeBernardi (2002) will be used to study perceived safety of women when train commuting in Metro Manila. The ABC Model explains that emotional and behavioral reactions are caused by beliefs regarding certain events rather than the external event itself. It creates a logical structure connecting the hypothesized relationships between identified variables using a psychological perspective. Modifications are made to explore more factors that may relate to perceived safety, which leads to a more robust conceptual model.

The results of this study will significantly benefit train operators and government agencies, aiding them in safety protocols and policies. It will also benefit participants by helping them understand the role of various activating events to their perceived safety. Finally, the results of this study shall help the government and NGOs in assessing its milestones in the 2030 SDG 11 goals, defined as "providing access to safe, affordable, accessible and sustainable transport systems for all, improving road safety, notably by expanding public transport, with special attention to the needs of those in vulnerable situations, women, children, persons with disabilities and older persons" (International Centre for the Study of the Preservation and Restoration of Cultural Property, n.d.).

2. Literature review

2.1. Women's safety

Women's safety from violent crimes has historically been a concern (Pain, 1997). Aside from the fear of being victims of crimes such as rape and murder, women are also unfairly given the burden to protect themselves against perpetrators when they go in public spaces (Valentine, 1989). Unfortunately, many women victims have suffered from both the consequences of crimes and the blame for putting themselves in situations of vulnerability. Although statistics show that women are more at risk in private places and from men they know, the fear has extended to the public space where there are more unfamiliar people, unwanted interactions, and uncontrollable factors (Valentine, 1989; Pain, 1997).

One particular domain of public space that continues to hound women's perceived and actual safety is public transportation. Public transportation is identified as one of the biggest public safety concerns for women (Day et al., 2003; as cited in Yavuz and Welch, 2010). This is evident in the differing mobility preferences and habits between men and women (Basbas et al., 2023). On top of that, extraordinary events can also add more difficulty for women to use public transportation. In rural India, the COVID-19 pandemic resulted in a lack of transportation options for women that negatively affected their education, socialization, and overall wellbeing (Tripathi, 2024).

2.2. Perceived safety of women in trains

Among the various modes of public transportation, train transit garners one of the greatest perceived safety risk, given the high volume of passengers in a strictly enclosed space with limited exit options (Walsh, 1999; Lynch and Atkins, 1988; as cited in Yavuz and Welch, 2010). Ouali et al. (2020) found that perceptions of metro characteristics significantly impact women's perceived safety. Specifically, their findings showed compelling evidence that the following factors had great influence: (1) total staff in stations, (2) the number of violent acts, and (3) the number of robberies, among others. Similarly, Bergdahl (2019) found that the social and physical environment influenced women's perceived safety on their public transportation journeys.

Yavuz and Welch (2010) likewise investigated the relationship of external factors and perceived safety in train transit environments. Their findings showed that external factors, such as the presence of security measures, influence perceived safety. Notably, women's perceived safety were more sensitive to these external factors compared to men, suggesting that their sense of safety is more deeply influenced by emotional and social cues. The study looked into relevant external factors that influence perceived safety, but it failed to account for psychological mechanisms that drive these perceptions. This gap is significant because while external factors like security measures are essential, without addressing the beliefs and emotions that contribute to perceived safety, interventions may fail to adequately address the root cause of women's fear in public transit environments. Understanding these psychological mechanisms is key to developing more effective interventions that shape especially women's overall sense of safety.

2.3. Psychological and environmental predictors of perceived safety

Media exposure represents a powerful community influence that shapes beliefs on a broader scale. It has been found to shape individuals' beliefs about danger and hazards (Reavley et al., 2016). Media consumption has stimulus control which can evoke immediate fear (Paudel, 2020). Furthermore, studies show that perceived safety tends to increase with repeated media coverage of threats, reinforcing fear and insecurity (Van der Pligt, 2001). Syam et al. (2011) further highlight the role of media exposure in shaping perceptions of safety specifically in public transport environments, where shared attitudes and social narratives significantly impact how individuals perceive their safety.

Aside from media exposure, it was also found by Gruber et al. (2021) that other people's views and testimonies heavily influenced an individual's beliefs. For instance, family members and close social networks play a crucial role in shaping and reinforcing negative social attitudes and stereotypes (Bryan et al., 1986; Pirchio et al., 2018; Zagrean et al., 2022). Taylor et al. (2006) suggest that growing up in a risky family environment is linked to challenges in developing effective emotion regulation skills. Discussions about dangers and safety concerns within these networks can heighten an individual's sense of risk, thereby impacting their perceptions of safety (Le Goff et al., 2023).

Beyond community-driven perceptions, women's perceived safety is significantly impacted by negative experiences with safety. Personal encounters with violence, hostility, and other negative life events are closely associated with heightened risk perceptions and mobility in general (Sagaris and Baker, 2024). People who have experienced gender-based violence are particularly vulnerable to detrimental mental and physical health effects (Sagaris and Baker, 2024; Sinko et al., 2022), as well as negative effects to overall life satisfaction (Alfaro et al., 2024). Individuals who have faced such experiences are more likely to perceive themselves as vulnerable and to believe that similar negative events are more likely to recur in the future (Blum et al., 2014).

Maran et al. (2022) and Sgritta (2023) revealed that witnessing sexual harassment—an indirect experience—led to significant emotional stress among witnesses. In particular, women's fear of future violence and sense of safety were influenced by both direct experiences and by witnessing or hearing about instances of violence. Structural vulnerability arises from observing marginalization (Casey et al., 2016). Those who had not personally experienced crime were impacted by the knowledge of crime in their transit environments, which heightened their fear and led them to adopt defensive behaviors (Kash, 2019; Akbolat et al., 2021). Unfortunately, data on incidences of sexual violence and harassment on public transport are not readily available, as well as among developed countries (Hooker et al., 2024).

2.4. The ABC model

The ABC Model (Wirga and DeBernardi, 2002) offers a framework for understanding how psychological responses interact to influence perceptions of safety. The model consists of three elements: Activating Event (A), Belief (B), and Consequences (C). It illustrates how external events (A) shape an individual's belief system (B), which in turn influences their emotional and behavioral responses or perceptions (C). This framework emphasizes that perceptions are not directly caused by external events, but rather by the individual's beliefs about those events.

Thus, the application of the ABC model in the context of women's perceived safety in train transit environments offers a more in-depth understanding of how safety perceptions are formed. The ABC model explores the underlying psychological processes that shape vulnerability. It emphasizes how women's beliefs and perceptions, in the form of perceived safety, are shaped and reinforced by external factors. By applying the ABC model, it becomes evident that interventions aimed at changing women's beliefs and perceptions of external threats, and still targeting the threats themselves, can be effective to increase feelings of security while also preventing violence against women. That is to improve both actual and perceived safety. For example, reshaping beliefs about the effectiveness of visible security measures could lead to women feeling more confident and secure when navigating public spaces. Ultimately, this approach is more comprehensive, integrating both psychological and environmental factors—something that Yavuz and Welch (2010) did not fully dissect in their focus on external influences.

Addressing this gap begins with identifying the pertinent activating events that potentially trigger women's perceptions of vulnerability in train transit environments. By definition, activating events refer to any occurrences or behaviors that an individual experiences or exhibits (Şahin and Voltan Acar, 2019). According to Ellis (1991), this concept also includes events that individuals contemplate but have not directly experienced. Using this definition, media exposure, familial upbringing, witnessing and experiencing negative experiences with safety can serve as activating events because they are external events that have an impact on an individual's belief (Paudel, 2020; Taylor et al., 2006; Sinko et al., 2022; Casey et al., 2016).

Beliefs refer to how a person reacts to the activating events (Şahin and Voltan Acar, 2019). It is the meaning that a person ascribes to their experience to the activating events. Beliefs play the primary role in the formation of consequences. Perceived safety acts as the belief of how safe women feel in their train commuting experience. The influence of the aforementioned activating events is channeled through the individual's subjective sense of risk or exposure.

Consequences are defined as the behavioral reaction or consequence that comes from beliefs (Şahin and Voltan Acar, 2019). Consequences may manifest as emotional responses, such as feelings of sadness, unhappiness, anxiety, or depression, or as behavioral patterns, like persistence in pursuing an activity or engaging in repetitive actions. Usage of train systems is the consequence that arises from the women's beliefs. This reflects how safe they feel in using such a form of public transportation.

This study will utilize a modified version of Wirga and DeBernardi (2002). The ABC Model will be applied to the study in order to supplement the analysis of the behavior of women in their train commuting experience. This would allow for better nuances in public policies and actions that make it more effectively address the key concerns of women's perceived safety. Fig. 1 shows the proposed

conceptual framework of the study. The framework proposes that activating events directly affect perceived safety, which then plays a key role as the belief that affects the women's actual usage of train systems.

2.5. The present study

Overall, the literature reviewed highlights how familial upbringing, media exposure, experiencing negative experiences, and witnessing negative experiences influence perceived safety. As such, the primary aim of this study is to understand the influence of psychological and environmental factors on the perceived safety of women train commuters in Metro Manila. Specifically, this study aims to test the following hypotheses:

H1. Women commuters' media exposure predict their perceived safety when using train services for commuting.

H2. Women commuters' familial upbringing predict their perceived safety when using train services for commuting.

H3. Women commuters' exposure to witnessing negative experiences with safety predict their perceived safety when using train services for commuting.

H4. Women commuters' exposure to experiencing negative experiences with safety predict their perceived safety when using train services for commuting.

3. Methods

3.1. Participants

The study included 200 women, aged 18 and above, residing in Metro Manila, Philippines, where the train systems in the country are concentrated. All participants must have used the Light Rail Transit (LRT), Metro Rail Transit (MRT), or Philippine National Railways (PNR) at least once within the past year. Table 1 below shows the frequency distribution of the annual household income of the participants.

The sample size was calculated using Green's (1991) Rule of Thumb for Regression Analysis that states $N \geq 50 + 8m$ for the multiple regression and $N \geq 104 + m$ for the simple regression where N indicates the number of subjects and m indicates the number of predictors. With this, 200 participants are more than sufficient to do a multiple regression analysis on Media Exposure, Familial Upbringing, Witnessing Negative Experiences with Safety, Experiencing Negative Experiences with Safety to Perceived Safety. Thus, for this multiple regression $N \geq 50 + 8*4$ since there are 4 predictors. Hence 200 respondents satisfy $N \geq 82$.

3.2. Materials

The study used a nonexperimental survey research design. The researchers collected data from participants through an online survey hosted on Google Forms.

The questionnaire had three sections:

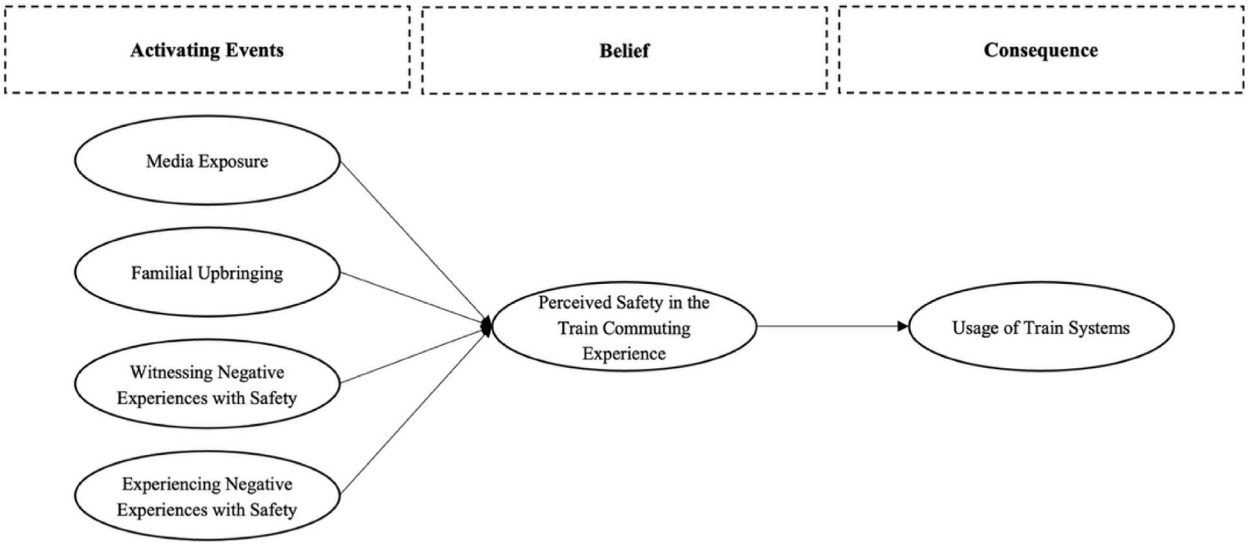

Fig. 1. The proposed conceptual framework.

Table 1
Frequency distribution of annual household income.

	freq	%
PhP 250,000 (approx. USD 4300) and below	71	35.5
PhP 250,000 to 400,000 (USD 4300 to 6900)	40	20.0
PhP 400,000 to 800,000 (USD 6900 to 13,700)	37	18.5
PhP 800,000 to 2,000,000 (USD 13,700 to 34,400)	36	18.0
PhP 2,000,000 (USD 34,400) and above	16	8.0

(1) an informed consent section;
(2) a section on demographic and background information; and
(3) a section where participants rate the extent to which they agree or disagree with the statements using a 4-point Likert scale.

3.2.1. Media exposure (ME)

ME measures how participants act in accordance with their received media exposure regarding the train system in the Philippines. Three questions were adopted from Alrasheed et al. et al. (2022) to assess its influence toward their perceived safety during train commuting in the Philippines (e.g., I depend on social media for information about the safety of the train system in the Philippines). The participants were asked to rate the statements using a 4-point Likert scale (ranging from 1 - “strongly disagree” to 4 - “strongly agree”).

3.2.2. Familial upbringing (FU)

FU measures the impact of the participants’ familial upbringing on their train commuting experience. Two questions were adopted from the study of Van den Berg et al. (2020) to assess the association between their familial upbringing and their perceived safety in the Philippine train system (e.g., My parents found it safe to allow me to ride the trains in the Philippines). The participants were asked to rate the statements using a 4-point Likert scale (ranging from 1 - “strongly disagree” to 4 - “strongly agree”).

3.2.3. Witnessing negative experiences with safety (WNS)

WNS measures the participants’ response to witnessing gender-based safety concerns while using the train system in the Philippines. Three questions were adopted from Lippincott’s (2020) Survey on the Patient Safety Incident Experience to measure the response (e.g., During the last years, I witnessed a severe and negative life incident that made me avoid the use of public trains or anything else associated with the event). The participants were asked to rate the statements using a 4-point Likert scale (ranging from 1 - “strongly disagree” to 4 - “strongly agree”).

3.2.4. Experiencing negative experiences with safety (ENS)

ENS measures the participants’ response to experiencing gender-based safety concerns while using the train system in the Philippines. Three questions were also adopted from Lippincott’s (2020) Survey on the Patient Safety Incident Experience to measure the response (e.g., It is still very painful every time I see places, things, or people reminding me of a negative incident I experienced). The participants were asked to rate the statements using a 4-point Likert scale (ranging from 1 - “strongly disagree” to 4 - “strongly agree”).

3.2.5. Perceived safety of women in the train commuting experience (PS)

PS measures the perceptions of the general safety of women when riding public trains in the Philippines. The researchers used the multimodal commuting stress scale from the study of Useche et al. (2023) to formulate three questions (e.g., I feel vulnerable to being harassed by other people during my commuting trips). The participants were asked to rate the statements using a 4-point Likert scale (ranging from 1 - “strongly disagree” to 4 - “strongly agree”). Given the phrasing of the items, the responses were reverse coded prior to data analysis.

3.3. Data collection procedure

Prior to the release of the questionnaire, a pretest was conducted where a sample of ten people were invited to provide qualitative feedback on the questionnaire’s clarity and relevance. The researchers conducted one-on-one interviews through Google Meet. For each interview, the participant was first briefed with the objectives and purpose of the study, emphasizing the importance of honest feedback. Participants were assured of confidentiality then were asked to complete the questionnaire as if they were actual respondents.

Afterward, researchers asked qualitative questions to gather feedback on clarity, relevance, and usability. Participants commonly highlighted that the questions were too broad and suggested specifying the contexts more clearly. This feedback, along with other suggestions on wording, was documented and used to refine the questionnaire. Consequently, some modifications were made to the adopted questions from previous studies to ensure greater clarity and alignment with the study’s goals.

After finalizing the research instrument based on the pre-test, the researchers recruited women commuters from Metro Manila

using purposive sampling. Recruitment was conducted primarily through social media, where the researchers joined Facebook groups such as "PAANO PUMUNTA: A Community Guide for Commuters PH," "The Manila Commuter FB Group," and "COMMUTERS @ BYAHEROS GROUP™." In these groups, the researchers posted the survey, inviting members who met the criteria to participate.

Additionally, personal connections were utilized to recruit respondents. To ensure eligibility, participants were required to confirm that they had used public train transit at least once in the past year. This confirmation was part of a brief preliminary screening included in the second section of the questionnaire, following the individual consent form.

3.4. Data analysis procedure

To test the model, regression analysis will be used. Multiple regression will determine whether the independent variables – Media Exposure (ME), Familial Upbringing (FU), Witnessing Negative Experiences with Safety (WNS), and Experiencing Negative Experiences with Safety (ENS) - significantly affect the dependent variable, Perceived Safety (PS).

4. Results

Table 2 below shows a summary of the descriptive statistics of the factors measured from the participants. Fig. 2 and Table 3 summarizes the results of the regression analysis.

The regression model predicting perceived safety (PS) shows that media exposure, familial upbringing, witnessing negative experiences with safety, and experiencing negative experiences with safety account for only 18.8 % of the variance in PS ($R^2 = 0.188$). The relatively low proportion of explained variance highlights the multifaceted nature of safety perceptions, suggesting that additional factors likely contribute to the outcome.

Media Exposure (ME) is a statistically significant predictor of Perceived Safety (PS), with a p-value of 0.001. The negative coefficient indicates that higher exposure to negative media content decreases PS.

Familial Upbringing (FU) is a statistically significant predictor of Perceived Safety (PS), with a p-value of 0.004. The positive coefficient indicates that as the level of familial upbringing involving positive discussions with the safety of train transit increases, there is an increase in PS.

Witnessing Negative Experiences with Safety (WNS) is not a statistically significant predictor of Perceived Safety (PS), with a p-value of 0.081. The negative coefficient suggests that, in theory, as the level of WNS increases, there could be a decrease in PS. However, this relationship is not supported by the data, and therefore, no conclusions can be drawn about the role of WNS in influencing women's PS in train commuting.

Experiencing Negative Experiences with Safety (ENS) is a statistically significant predictor of Perceived Safety (PS), with a p-value of 0.032. Taking into account the negative coefficient, higher experiences with negative safety incidents is thus associated with lower PS during train commutes.

5. Discussion

The results reveal a high level of media exposure on the train commuting experience among female commuters. This indicates that media plays an important role in shaping their knowledge and perceptions about the train commuting experience. Similarly, there was also a high level of familial upbringing (FU) on female commuters' perceptions of the train commuting experience.

In contrast, the results reveal a low level of exposure on both witnessing and experiencing negative experiences with safety (WNS and ENS) among female commuters, given a mean below 2.5. This means that on average, women disagree that they have witnessed or experienced negative experiences that gave them emotional distress.

The results also reveal a low level of perceived safety (PS) among female commuters, highlighting a lack of sense of security and widespread feelings of vulnerability during the train commuting experience.

In terms of the model, the findings indicate varying levels of significance across the predictors, underscoring the complexity of predicting perceived safety among female commuters.

Media Exposure (ME) emerged as a significant predictor of perceived safety, supporting H1. The positive slope of ME suggests that increased exposure to negative media content leads to heightened perceptions of vulnerability. This finding aligns with Paudel (2020), who reported that excessive media consumption can amplify fear through stimulus control. The results highlight the influence of media narratives on public perceptions of safety.

Similarly, Familial Upbringing (FU) was a statistically significant predictor of perceived safety, supporting H2. This finding

Table 2
Summary of descriptive statistics.

	M	SD
Media Exposure	3.47	0.54
Familial Upbringing	2.84	0.96
Witnessing Negative Experiences with Safety	1.99	0.82
Experiencing Negative Experiences with Safety	2.09	0.85
Perceived Safety	1.93	0.75

Table 3
Regression analysis of factors influencing the perceived safety (PS).

	Unstandardized Coefficient B	Standardized Coefficient B	t	p-value
ME	−0.32	−0.23	−3.45	0.001
FU	0.15	0.19	2.89	0.004
WNS	−0.14	−0.15	−1.76	0.081
ENS	−0.16	−0.18	−2.16	0.032

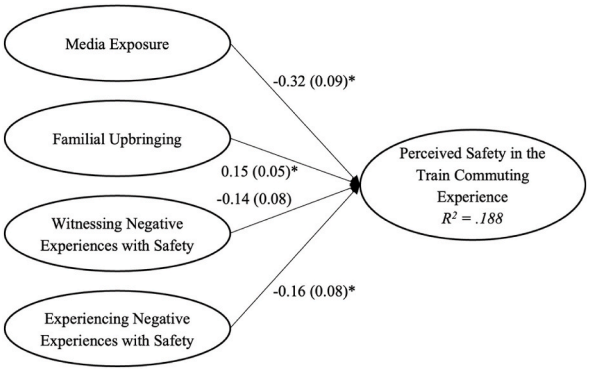

Fig. 2. Structural model with unstandardized Coefficients
Note: *p < .05.

supports existing literature, such as Gruber et al. (2021), which highlights the prominent influence of others' views and testimonies on individual beliefs. Family and social networks are known to shape negative attitudes and amplify safety concerns, heightening perceptions of risk and vulnerability (Bryan et al., 1986; Pirchio et al., 2018; Zagrean et al., 2022; Le Goff et al., 2023).

Experiencing Negative Experiences with Safety (ENS) is also a significant predictor in the model, supporting H4. This aligns with the findings of Chowdhury and Van Wee (2020), who reported that women's experiences with gender-based safety concerns during commutes often heighten their vulnerability, leading to increased fear of victimization.

In contrast, Witnessing Negative Experiences with Safety (WNS) was not a significant predictor of perceived safety, failing to supporting H3. This result is inconsistent with the findings of Maran et al. (2022) and Sgritta (2023), who noted that witnessing or hearing about unsafe situations can evoke emotional stress and fears of future violence. This may suggest that witnessing negative experiences does not worsen women's perceived safety further, perhaps because their perceived safety in train commuting is already high. This may also be due to a possible collinearity with Experiencing Negative Experiences with Safety (ENS), which was a significant predictor of perceived safety.

Among all the predictors of Perceived Safety, Media Exposure emerges as the most significant predictor with a standardized coefficient β equal to −0.23, followed closely by Familial Upbringing (β = 0.19), and Experiencing Negative Experiences with Safety (β = −0.18).

The results highlight the complexity of factors influencing perceived safety (PS) among women in train transit environments. FU, ME, WNS, and ENS were examined as predictors of PS. While the model demonstrated moderate fit, explaining 18.8 % of the variance in PS, the limited explanatory power suggests that additional factors may play a critical role. Future research can consider other factors, such as institutional support, cultural norms, and environmental factors. The limited explanatory power may also be influenced by the use of Likert-scale measures, which can introduce biases and potentially underestimate the model's effectiveness. As Owuor (2001) noted, Likert-scale biases, such as restricted variance and respondent subjectivity, can affect the reliability of predictive models.

The study conducted by Yavuz and Welch (2010) emphasizes that one of the biggest concerns regarding public train transit safety is the perceived safety, especially among women. This study affirms this concern, revealing that women in the Philippines indeed exhibit a low level of perceived safety when using the train commuting system. Such perceptions have direct consequences on their use of public transportation, often negatively impacting their commuting experience. Given that safety is a fundamental element in societal well-being, it is crucial to evaluate factors that influence safety perceptions. A comprehensive understanding of the impact of these factors is necessary to foster safer and more inclusive public spaces.

The findings of this study have significant implications for addressing safety concerns in train transit. Media Exposure (ME) emerged as a significant predictor of perceived safety, confirming that increased exposure to negative media content amplifies perceptions of risk, a finding consistent with Paudel (2020). This highlights the need to ensure that legitimate security concerns in public spaces are broadcasted to the public while guarding against media misrepresentation that aims to invoke fear among women and manipulate their behavior. Moreover, Familial Upbringing (FU) was a statistically significant predictor of perceived safety, suggesting the important role that families play in assuring women's safety when they take public transportation. Lastly, Experiencing Negative

Experiences with Safety (ENS) was also a significant predictor of perceived safety, aligning with the work of Maran et al. (2022) and Chowdhury and Van Wee (2020). These findings stress the importance of addressing actual security incidents of the public, as experiencing safety violations can heighten women's fears and vulnerability in public transit environments.

Ultimately, the findings suggest that the most effective way to increase women's perception of safety in train commuting is to reduce the actual frequency and severity of security incidences in the train systems. By doing so, Experiencing Negative Experiences with Safety (ENS) will be decreased, Media Exposure (ME) will be decreased, and Familial Upbringing (FU) will be increased, which will all lead to an increase in Perceived Safety (PS). Concretely, this can be done through various measures that have been suggested in the literature. More security personnel could be assigned at various locations in the train systems, such as in the hallways, platforms, and inside the trains themselves. These personnel should also be visible at all times while the trains are operating, even during off-peak hours where the number of passengers are fewer. More than staff visibility, the management should also increase the likelihood that perpetrators of crimes in these places will be caught and punished. When there is high certainty of punishment among perpetrators of violence in train systems, people will be deterred to commit crimes and negative security incidences will be reduced.

Despite the valuable insights provided, this study has notable limitations. The sample was limited to residents of Metro Manila, which may limit the generalizability of the findings. The findings may only be applicable to areas with similar demographics like the participants of the study, public transportation infrastructure, as well as crime rates, such as other cities in Southeast Asia. Future studies should expand the participant pool to include commuters from other nearby regions to obtain a more representative understanding of the issue. Additionally, the limitations of the statistical method used in this study suggest that future research employing more advanced analyses like structural equation modeling would be beneficial. Future studies may also consider non-linear models or interaction effects to better capture the intricate relationships between predictors and perceived safety, ultimately leading to more robust findings and interventions to improve safety in public transit systems.

Furthermore, the low explanatory power of the current model highlights the need for future research to incorporate additional factors that may better capture the complexities of perceived safety. The current study failed to account for the possible influence of limited public transportation alternatives, affordability, and convenience in predicting perceived safety. As such, exploring psychological, environmental, and socioeconomic factors could provide a more comprehensive understanding of safety perceptions. Moreover, the current study was only able to test the hypotheses between activating events and belief in the ABC Model. Future studies could measure actual usage of trains for commuting in order to establish the connection between the belief and the consequence in the ABC Model. Future studies may also look conduct a discourse analysis of security incidents reported in media to analyze their accuracy and representation of public transport.

CRediT authorship contribution statement

Claire Louisse L. Basallo: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Joseph Andrei S. Carlos:** Writing – original draft, Methodology, Formal analysis, Conceptualization. **Mikayla Rae G. Guerrero:** Writing – original draft, Methodology, Formal analysis, Conceptualization. **Airazelle Sheerie M. Malaluan:** Writing – original draft, Methodology, Formal analysis, Conceptualization. **Stephen Daeniel Mansueto:** Writing – original draft, Methodology, Formal analysis, Conceptualization. **Alvin Patrick M. Valentin:** Writing – review & editing, Validation, Supervision, Formal analysis, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

APPENDIX A

Media Exposure

1. I use the internet to update myself with news about the local community.
2. I depend on social media for information about the safety of the train system in the Philippines.
3. I like to read about other people's opinions about certain topics that arise in the community.

Familial Upbringing

1. My parents found it safe to allow me to ride the trains in the Philippines.
2. As a child, my parents encouraged me to learn how to ride the public trains alone.

Witnessing Negative Experiences with Safety

1. During the last years, I witnessed a severe and negative life incident that made me avoid the use of public trains or anything else associated with the event.
2. During the last years, I witnessed a severe and negative life incident that made me feel helpless and disempowered.
3. After witnessing the incident, I was very much afraid of experiencing a similar incident.

Experiencing Negative Experiences with Safety

1. It is still very painful every time I see places, things, or people reminding me of a negative incident I experienced.
2. I still live with the impacts of the incidents.
3. Although a long time has passed, I am often still distressed by the painful memories of the incident.

Perceived Safety

1. When I commute, I feel vulnerable to being robbed by other people.
2. I feel vulnerable to being harassed by other people (e.g., words, looks, groping …) during my commuting trips.
3. Physically, I feel a bit tense while commuting.

Data availability

The authors do not have permission to share data.

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